What is DevOps?

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Definition

DevOps is a methodology that creates a connection between **software developers (Dev)** and **IT operations teams (Ops)**.

The word **DevOps** comes from:

DEV = Development
→ Creating and developing software applications.

OPS = Operations
→ Managing systems, servers, networks and IT infrastructures.

The main goal of DevOps is to help teams **build, test, deploy and maintain applications faster and more efficiently**.

DevOps in simple words

Before DevOps:

👨‍💻 Developers create an application.

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They send their work to IT operations teams.

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The operations team installs and manages the application.

This process could create problems:

- The application does not work correctly on the server.
- Software versions are not compatible.
- Teams do not communicate efficiently.

With DevOps:

👨‍💻 Developers and IT operations teams work together.

They use tools and processes to:

- automate tasks;

- test applications automatically;
- deploy new versions faster;
- monitor application performance.

What do students learn in a DevOps training program?

1. IT Systems Fundamentals

Students learn the basics of:

- **Servers**
- **Linux systems**
- **Networks**
- **Databases**
- **Cloud environments**

Example:

The application runs on a Linux server.

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2. Version Control with Git

Git is a tool that allows teams to work together on software projects.

Important vocabulary:

English	Meaning
Repository	A place where code is stored
Commit	Save a change in the code
Branch	A different version of the code
Merge	Combine different versions of code

Example:

I pushed my code to the Git repository.

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3. Automation

A key principle of DevOps is to automate repetitive tasks.

Teams use automation to:

- run tests;
- install applications;
- deploy software;
- manage systems.

Example:

The deployment process is automated.

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4. CI/CD (Continuous Integration / Continuous Deployment)

CI/CD is an important concept in DevOps.

Continuous Integration (CI)
means regularly adding and testing code changes.

Continuous Deployment (CD)
means automatically releasing new versions of an application.

Example:

A developer changes the code
→ The system runs tests
→ The application is deployed automatically.

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5. Containers (Docker)

Docker is a technology used to create containers.

A container contains everything an application needs to run correctly.

Important vocabulary:

English	Meaning
Container	A package containing an application and its environment
Image	A template used to create containers
Deploy a container	Put a container into operation

Example:

Docker allows developers to package applications.

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6. Cloud Computing

DevOps professionals often work with cloud platforms such as:

- AWS
- Microsoft Azure
- Google Cloud

Cloud computing allows companies to use IT resources remotely.

Example:

The application is hosted in the cloud.

Why is English important in DevOps?

English is essential because most technical resources are written in English:

✓ Technical documentation
✓ Online tutorials
✓ Developer forums
✓ Software tools
✓ Error messages
✓ International meetings

Common DevOps vocabulary:

English	French
server	serveur
deployment	déploiement
error	erreur
issue	problème
bug	anomalie
update	mise à jour
backup	sauvegarde
security	sécurité
database	base de données
pipeline	chaîne automatisée

Professional English Objective

In this course, the objective is not to teach DevOps concepts.

The objective is to learn **how to communicate in English in a DevOps environment**.

You will learn how to:

- understand technical documentation;
- participate in professional meetings;
- write emails and tickets;
- explain technical problems;
- communicate with international teams.

English is a key skill for DevOps professionals because technology is global !